BOEING CASE COMPETITION

Strengthening Boeing's Supply Chain

Kappa Theta Pi Consulting

BOEING

CASE COMPETITION



Ryan Kauch
Computer Science



Teagan Donnelly
Computer Science



Tristian Peyton
Computer Science



Rasail Shrestha
Electrical Engineering

Executive Summary

BOEING

Goal: Strengthen Boeing's supply chain by developing qualification programming, expanding manufacturing capacity, and securing critical components.

Education

Train skilled workers for critical Tier 2/3 manufacturers

Partnerships

Secure supplier capacity through long-term agreements and investments

Fabrication

Qualify 3D-printed components to diversify production

Key Performance Indicators		
Skilled workers trained & placed	Strategic supplier agreements	Alternative parts qualified
Tier 2/3 hiring needs met	Long-term capacity secured	FAA qualification achieved

Agenda

BOEING



Analysis

Supply Base & Leverage Gap

1



Strategy

Strategy & Phased Plan

2



Execution

Execution Plan & Timeline

3



Results

Cost & KPIs

4



Risks

Risks & Mitigation

5

Our Understanding of Boeing's Needs

BOEING

Internal Factors:

- Limited visibility into Tier 2/3 suppliers
- Extended supplier payment terms (up to 120 days)
- Lengthy qualification process for alternative suppliers

External Factors:

Private Equity Consolidation
+25.8% PE add-on acquisitions

Skilled Labor Shortages
56% of aerospace suppliers affected

Manufacturing Capacity Constraints
Nearly 40% report insufficient capacity

Rising Raw Material Costs
Steel +27.6% YoY

Data Center Competition
$720–745B in 2026 hyperscaler spending

Sources: Capstone Partners, *Commercial Aerospace M&A Update* (2026); Roland Berger, *Global Aerospace Supply Chain* (2026); BLS, *Producer Price Index* (Sept. 2026); TMT Finance, *Hyperscaler Capex Analysis* (Sept. 2026).

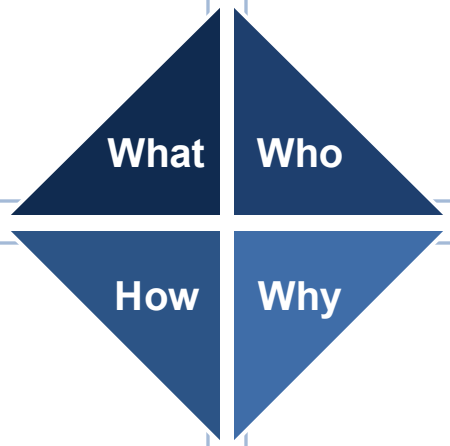
Current Environment: Boeing's Supply Base

What is happening

- PE ownership is expanding: Arcline acquired Kaman for ~$1.8B (April 2024)
- Worker, material and part shortages slow how fast suppliers can grow
- 74% of A&D executives say skilled trades are their biggest labor constraint

Who holds leverage

- PE firms that own qualified Tier 2/3 suppliers
- Arcline: Arxis (Kaman, Schatz Bearing) and Novaria; Avantus pending
- Boeing is left with fewer alternative suppliers



How leverage is created

- Boeing D1-4426 limits approved sources for special processes and materials
- Fewer independent owners means fewer options in negotiations
- Only 37% have visibility into their external supply chain

Why now

- Record $612B commercial backlog, but Q3 deliveries rose only 1%
- Suppliers want firm orders before adding capacity
- Current efforts are small: Ascend supports 10–15 suppliers

Sources: AIA/PwC, State of the Ramp (Jul 2026); Business Wire, Arcline Completes Acquisition of Kaman (Apr 2024); Arcline and Arxis releases (Apr, Sep 2026); Inflexion (Jul 2026); Boeing Suppliers, D1-4426 (2024); Boeing Q3 2026 Results; UW Foster, Ascend (May 2026).

Root Causes of the Leverage Gap

BOEING

PE Consolidation

~1 in 3 A&D deals by PE

- One acquisition turns two qualified sources into one
- Years to qualify a replacement source
- Change-of-control clauses give limited practical leverage

Option to close the gap

- Training-led partnerships to build foundry and supply access
- FAA-qualified 3D printing as an alternate source

Expanding Competition

$700B+ data center capex

- 12 shared component categories
- Gas turbines and jet engines draw on the same superalloy foundries
- AI buyers offer prepayment vs. Boeing's ~120-day terms

Option to close the gap

- 501(c)(3) training pipeline builds recruiting leverage
- Digital twin to prioritize scarce supplier capacity

Declining Resources

$28.92 to $329.17 per MWD

- Five key metals: 27.6% increase
- Shortage of 150K A&D engineers
- 56% of suppliers report skilled-personnel shortages

Option to close the gap

- Reduced power usage
- Alternative materials
- Streamlined education

Sources: Boeing Case Booklet (2026); PJM, 2026/27 Auction Report; EIA, Electric Power Annual (2020, 2025); Capgemini (2026); Roland Berger (2026).

Strategic Options to Close The Gap

BOEING

Earlier Payment Incentives

Description

Offer earlier invoice payment or milestone advances for agreed deliveries.

Pros

Supports supplier cash flow.

Cons / Trade-offs

Uses Boeing cash without securing added capacity.

Selective Supplier Acquisition

Description

Acquire a qualified supplier tied to a critical part bottleneck.

Pros

Provides direct operational control.

Cons / Trade-offs

Requires acquisition funding and integration.

Training Nonprofit

Description

Prepare candidates for confirmed supplier jobs and support skills progression.

Pros

Builds a supplier hiring pipeline.

Cons / Trade-offs

Needs trainee retention and time to proficiency.

Supplier Capacity Partnerships

Description

Fund a verified bottleneck in exchange for agreed Boeing capacity.

Pros

Secures measurable delivery commitments.

Cons / Trade-offs

Needs supplier agreement and proof of added output.

3D Printing + Digital Twin

Description

Model production bottlenecks and qualify suitable printed parts.

Pros

Tests improvements and alternative production routes.

Cons / Trade-offs

Requires reliable data, testing and part qualification.

Evidence: AIA/PwC, State of the Ramp (2026); Boeing, Supplier Quality; Orion Industries; NIST, AM Qualification and Digital Twin Credibility.

Strategy Overview: FORGE

BOEING







Phase 1

Launch Boeing Education
Establish the organization and scholarship funding

Build Supplier Talent Pipeline
Partner with technical colleges and Tier 2/3 manufacturers

Phase 2

Build Workforce Partnerships
Match trained graduates with verified Tier 2/3 supplier openings.

Secure Production Commitments
Negotiate multiyear orders, reserved capacity, and delivery targets.

Phase 3

Deploy Supply Chain Digital Twin
Map critical Tier 2/3 suppliers to predict disruptions and guide investments.

Qualify 3D-Printed Alternatives
Develop certified backup components to reduce sole-source dependency.

Phase 1 – Launch Boeing Ed (Months 0–12)

BOEING

50	$500K	2
pilot trainees in year one	tuition budget, funded by Boeing	training tracks (CNC and inspection)

1	2	3	4	5	6	7
Establish Boeing Ed	**Identify critical suppliers**	**Build the curriculum**	**Partner with colleges**	**Fund scholarships**	**Train and place**	**Evaluate the pilot**
Nonprofit setup and Boeing funding	Find Tier 2/3 workforce gaps	CNC machining and quality inspection	Technical colleges near key suppliers	$500K tuition for 50 trainees	Place trainees at suppliers	Track completion and placement

Illustrative college partners: Dutchess Community College, CT State Asnuntuck, St. Louis Community College, Tarrant County College. Sources: manufacturer websites; technical college program catalogs; Rock Valley College CNC program.

Strategy Overview: FORGE

BOEING







Phase 1

Launch Boeing Education

Establish the organization and scholarship funding

Build Supplier Talent Pipeline

Partner with technical colleges and Tier 2/3 manufacturers

Phase 2

Build Workforce Partnerships

Match trained graduates with verified Tier 2/3 supplier openings.

Secure Production Commitments

Negotiate multiyear orders, reserved capacity, and delivery targets.

Phase 3

Deploy Supply Chain Digital Twin

Map critical Tier 2/3 suppliers to predict disruptions and guide investments.

Qualify 3D-Printed Alternatives

Develop certified backup components to reduce sole-source dependency.

Phase 2 – Partnerships (Year 2)

Workforce Partnerships

Turn Phase 1 trainees into supplier hires

What

Connect Phase 1 trainees to supplier-confirmed openings for entry-level machine operators

Supplier value

Job-ready candidates, with training and follow-up after placement

Boeing impact

Trade training support for hiring and capacity commitments; track hires, retention and output on Boeing parts

Production Agreements

Lock in capacity with multiyear orders

What

Multiyear purchase commitments in exchange for reserved capacity, delivery milestones and supplier reporting

Supplier value

Predictable orders give suppliers confidence to hire, buy materials and invest in equipment

Boeing impact

Enforceable capacity and delivery commitments; track on-time delivery, lead times and reserved capacity

Sources: RBC Bearings, Machine Operator job posting; Orion Industries, Employer Services; AIA/PwC, State of the Ramp (2026).

Strategy Overview: FORGE

BOEING



Phase 1

Launch Boeing Education

Establish the organization and scholarship funding

Build Supplier Talent Pipeline

Partner with technical colleges and Tier 2/3 manufacturer



Phase 2

Build Workforce Partnerships

Match trained graduates with verified Tier 2/3 supplier openings.

Secure Production Commitments

Negotiate multiyear orders, reserved capacity, and delivery targets.



Phase 3

Deploy Supply Chain Digital Twin

Map critical Tier 2/3 suppliers to predict disruptions and guide investments.

Qualify 3D-Printed Alternatives

Develop certified backup components to reduce sole-source dependency.

FORGE Digital Twin

The plan.

737 FINAL ASSEMBLY



Aircraft this month

42 on track

42 planned

At risk from missing parts

0 aircraft

Out-of-order work

0 hours

Parts to make in-house

0 + 0

now / after approval

A simulated view of the 737 line and the suppliers that feed it.

Unofficial student concept · All data simulated · Not affiliated with Boeing

Implementation: Owners & Governance

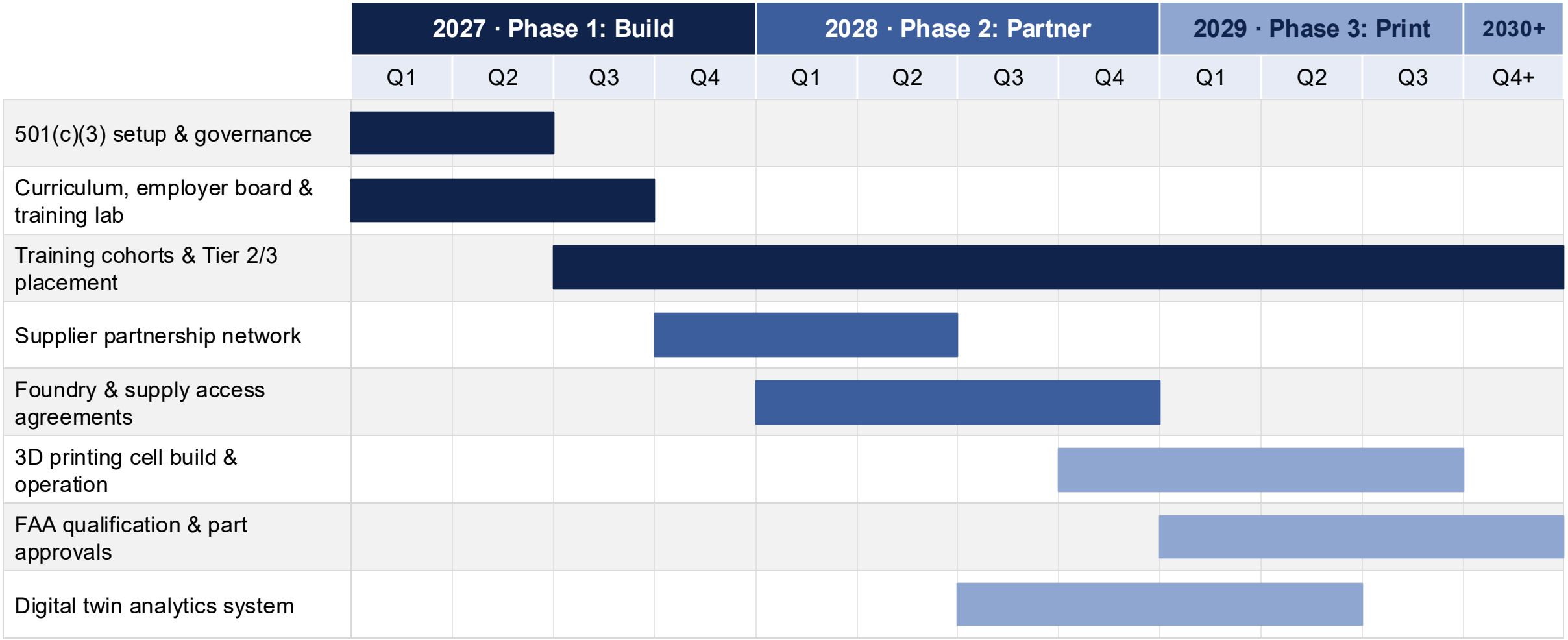
BOEING

Orion Industries, Employer Services; AIA/PwC, *State of the Ramp* (2026); NIST, AM Qualification; NIST, Digital Twins for Advanced Manufacturing; NIST, Digital Twin Credibility (2022)

Workstream	Owner	Governance / Decision Forum	Resources	Key Risk
Nonprofit	Boeing program director/board	Monthly nonprofit supplier review, approve cohorts against confirmed hiring needs	Instructors, training equipment, tuition funding, supplier mentors	Skills taught do not match supplier hiring requirements
Supplier production partnerships	Boeing procurement lead	Quarterly Boeing supplier steering committee, approve commitments and resolve delivery issues	Procurement, legal support, demand forecasts, funding where justified	Commitments lack funding needed to expand capacity
3D-printing partnerships	Boeing additive manufacturing lead	Engineering and quality stage reviews; approve part selection, testing, and production release	Technology partner, suitable equipment and materials, testing and qualification budget	Qualification requires more testing, time, or spending than planned
Digital twin implementation	Boeing manufacturing engineering lead	Monthly engineering supplier review; validate the model before approving operational changes	Simulation software, supplier production data, engineers, data integration support	Insufficiently validated model produces unreliable recommendations

Timeline

BOEING



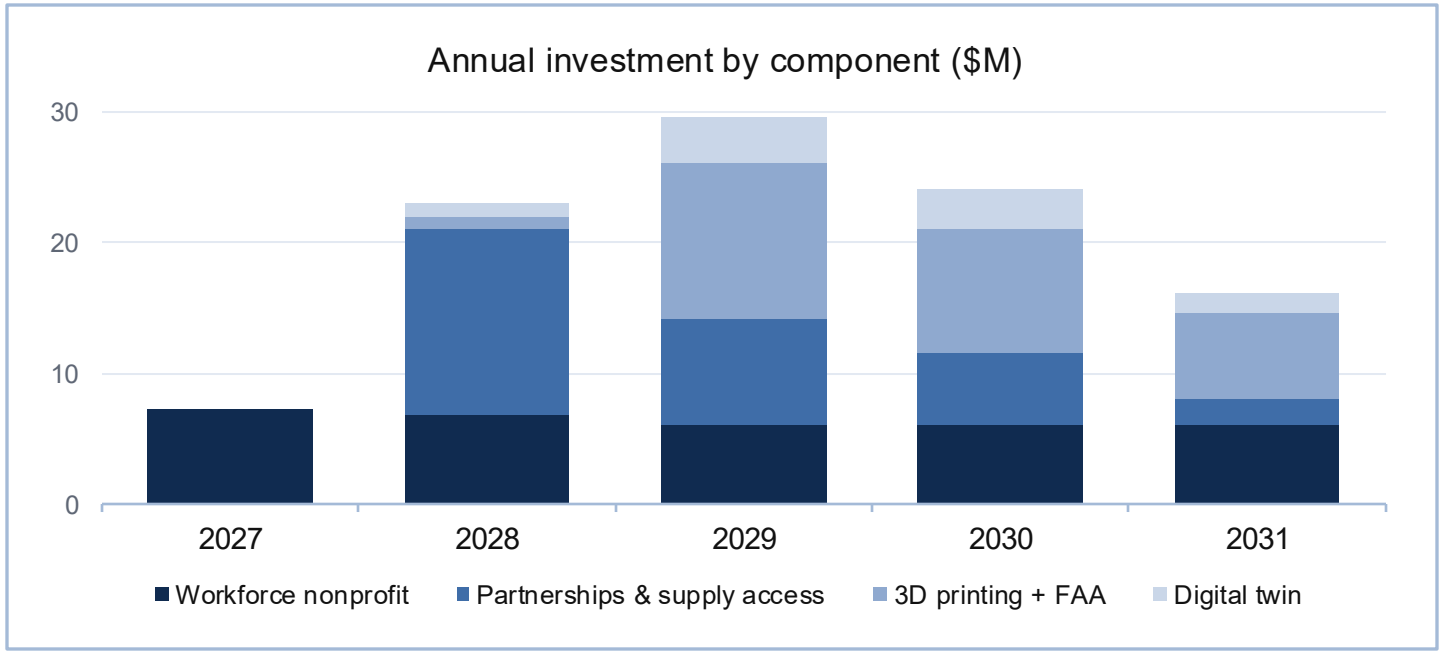
Cost of the Strategy

BOEING

≈$100M	+$9.8M	1.1x	≈$41M	2031
total investment, 2027–31	net 5-year value	return on investment	peak cumulative outlay	breakeven year

Key assumptions

- Training: $12K per trainee
- Partnerships: $18M co-investment
- 3D printing: ~$1.5M per metal printer
- Digital twin: $6M to build
- Benefits: 40% of spend re-sourced



Team financial model, 2027–2031. Totals by year: 2027 $7.2M · 2028 $23.0M · 2029 $29.6M · 2030 $24.1M · 2031 $16.1M.

Financial Impact & Sensitivity

BOEING

$500K
initial investment for 50 trainees

1,000
trainees planned over five years

$10M
five-year tuition (24-week program)

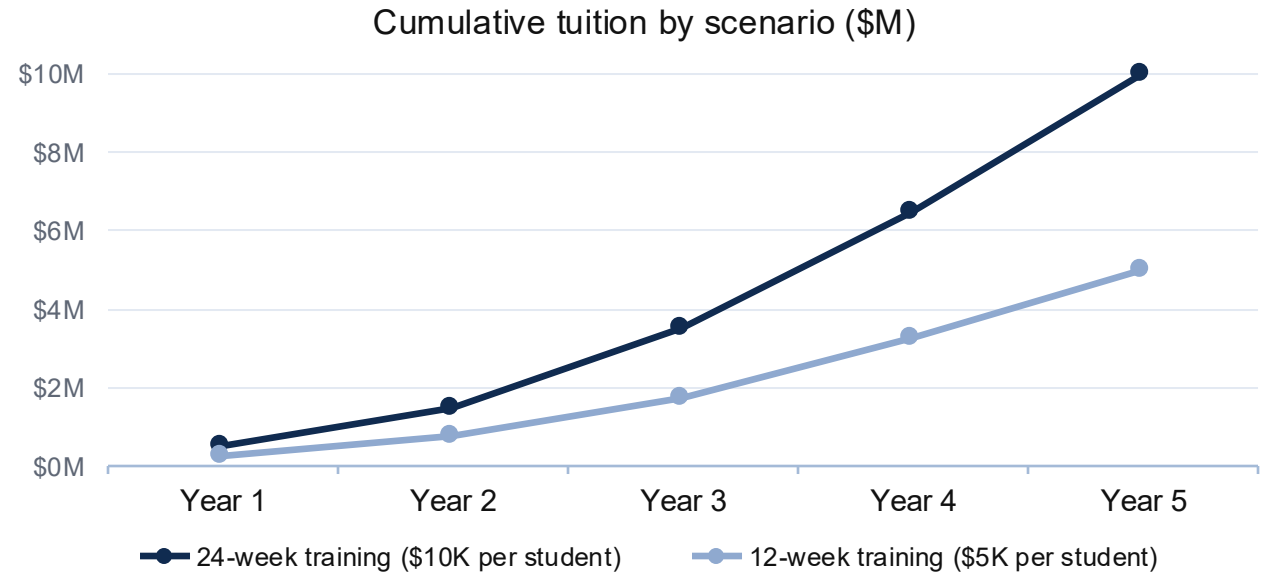
How we calculated it

Tuition: $10,000 per trainee for a 24-week CNC program

Five-year cost: 1,000 × $10,000 = $10M

Sensitivity: a 12-week program costs $5M instead of $10M

Impact: a stronger supplier workforce, capacity access and less disruption risk



Illustrative tuition-only projection; excludes program operations and administration. Basis: Rock Valley College CNC training benchmark; KTP Consulting enrollment assumptions.

Key Performance Indicators

KPI	Definition	Baseline	Target	Timing
Workforce Development	Students enrolled in Boeing Ed	750 workers	1,000 Trained	Year 1
Supplier Partnerships	New capacity agreements secured	2 agreements	3 agreements	Year 2
Alternative Sourcing	Backup part numbers qualified	1 part	2 parts	Year 5
Supply Chain Visibility	Priority Tier 2/3 suppliers mapped	70% mapped	90% mapped	Year 5

Risks & Mitigation

Risk	Description	Mitigation	Likelihood	Impact
Workforce Placement	Graduates may choose employers outside Boeing's supply chain.	Target verified vacancies; offer voluntary supplier recruiting pathways.	High	Medium
Supplier Commitment	PE-owned suppliers may reject capacity agreements or change priorities.	Offer multiyear orders and negotiate reserved capacity with ownership-change protections.	Medium	High
3D-Printing Qualification	Aerospace certification delays could prevent alternative parts from entering production.	Prioritize feasible parts; use phased testing and qualification milestones.	High	High
Digital Twin Data Quality	Limited Tier 2/3 visibility may undermine predictive accuracy.	Require agreed data sharing, validate supplier inputs, and pilot before scaling.	High	High

Thank you!

Any questions?

Directory: Model and Evidence

BOEING

Our model

40 – Annual Cost by Component ($M)
41 – Key Assumptions
42 – Headline Results
43 – Scale of the Plan vs Boeing Cash
44 – Cost vs Acquiring Capacity
45 – Cost of a Stoppage vs Our Plan
46 – Phase 1 Recap (Months 0–12)
47 – Phases 2 and 3 Recap
48 – Timeline Recap
49 – KPI Recap
50 – Governance Recap
51 – Risk Recap
52 – Reconcile Before Q&A

Evidence (1)

54 – FAA Approach to 3D-Printed Parts
55 – Certified 3D-Printed Aircraft Parts
56 – Cost, Time and Barriers of AM Qualification
57 – AM and Mechanical-Standards Parts
58 – Qualification Time for New Sources
59 – OEM Investment in Suppliers
60 – Boeing Spirit Acquisition and FTC Order
61 – GE Aerospace Buying CPP

Evidence (2)

62 – Sole-Source Pricing Power
63 – Antitrust Context
64 – Customer-of-Choice Concept
65 – Payment Terms and Supplier Finance
66 – Long-Term Supplier Agreements
67 – Machinist Wages and Demand
68 – Training Program Examples

Appendix A1: T-7A Red Hawk – Digital Engineering Promise and Program Timeline

What was promised

- Boeing's T-7A was designated eT-7A before delivery to mark it as a digitally engineered aircraft (Wikipedia, citing FlightGlobal, Sept. 2020).
- Boeing described it as a "fully digitally designed aircraft" built with advanced manufacturing, agile software development and digital engineering (The Defense Post, Apr. 2023).
- Developed "in fewer than three years" using digital design; designed with a digital thread aligned with the Air Force's Digital Century Series strategy (Air & Space Forces Magazine).
- "Went from development to the first test flight within 36 months" (Wikipedia, citing The Drive, Sept. 2019). The start point is not stated; the T-X prototype first flew Dec. 20, 2016, about 21 months before the Sept. 2018 award.
- Digitized production line: 30 minutes to splice the aft section to the wings, with minimal tooling and drilling during assembly (Wikipedia).

Program timeline

- Sept. 2018: Air Force selects Boeing; up to $9.2B for 351 aircraft, 46 simulators and support, with options up to 475 aircraft.
- Q3 2018: Boeing records a $691M pre-tax charge, in part because of the T-X program.
- Feb. 2021: Engineering and manufacturing development phase begins.
- Apr. 28, 2022: first T-7A rolled out.
- June 2023: first T-7A test flight with an Air Force pilot, from St. Louis.
- Nov. 9, 2023: first of five production-representative aircraft delivered to Edwards AFB.
- Dec. 5, 2025: first operational T-7A arrives at JBSA-Randolph (12th Flying Training Wing).
- May 2026: Air Force approves entry into initial low-rate production.

Appendix A2: T-7A Red Hawk – Schedule Delays, Technical Issues and Testing

BOEING

Schedule

- Jan. 2023 schedule judged “optimistic and dependent on favorable assumptions” (GAO, May 2023). Boeing gave a Milestone C date of “no later than February 2025”; initial operational capability was expected in 2027, nearly two years past original expectations.
- Jan. 16, 2025: Lot 1 production delayed to 2026, one year later than planned; FY2025 production cut from 14 to 7 airframes; four more production-representative test vehicles added with FY2025 R&D funds.
- Andrew Hunter, Assistant Secretary of the Air Force for Acquisition: “Acquisition programs cannot be stagnant, even when they are fixed-price.”
- Air & Space Forces Magazine: early-test deficiencies and supply chain problems delayed the low-rate production decision two years to 2026; it lists initial operational capability in 2028. Wikipedia cites 2027 (source conflict).
- July 2026 GAO (per Wikipedia): full-rate production decision slips two years, from 2027 to 2029, due to delayed developmental testing and additional engineering analysis.

Technical issues, testing and cost

- Escape system (ejection seat) cited as the reason for the Dec. 2022 delay and flagged by GAO as a top schedule risk. Reports cited concussion risk, unsafe acceleration at parachute opening and possible helmet visor loss.
- Boeing flight tests focused on type certification: instability at high angles of attack, ejection seats, and tuning of flight control software, engines and flight dynamics.
- Air Force developmental testing covers wing flutter, flight dynamics, structural loading and the onboard oxygen generation system.
- Air Force cited aerodynamic instability, hardware qualification challenges, subcontractor design delays and slow correction of deficiencies (2023).
- Frank Kendall, Air Force Secretary (May 23, 2023 headline): digital engineering was “Over-Hyped,” but can save 20 percent on time and cost. Article text not reviewed.
- Boeing charges: headlines report $1B on VC-25B and T-7A (Apr. 2022) and $222M on T-7 and KC-46 (Apr. 2024); per-program amounts not verified.

Sources: Air Data News (Jan. 16, 2025); The Defense Post (Apr. 24, 2023); Air & Space Forces Magazine, T-7A Red Hawk; Wikipedia, Boeing–Saab T-7 Red Hawk (cites GAO). Wikipedia is a secondary source.

Appendix A3: Boeing's 2021 Digital Strategy – Promise, Supplier Barriers and Critiques

The promise (Reuters interview, Dec. 17, 2021)

- Chief engineer Greg Hyslop described a plan to unify design, production and airline services under one digital ecosystem "in as little as two years."
- Goal: linked virtual 3-D "digital twin" replicas of the jet and the production system, plus a "digital thread" tying together airline requirements, millions of parts and thousands of pages of certification documents, "extending deep into the supply chain."
- Factory tools: immersive 3-D designs twinned with robots; mechanics linked by $3,500 Microsoft HoloLens headsets.
- Hyslop: "More than 70% of quality issues at Boeing trace back to some kind of design issue." Boeing believes the tools could bring a new aircraft to market in as little as four or five years.
- Linda Hapgood tapped to oversee the "digital transformation," reportedly underpinned by more than 100 engineers (industry source).
- Context: groundwork for a future aircraft program over the next decade, described as a $15 billion gamble.

Supplier barriers and critiques

- "Hundreds of smaller suppliers spread globally lack the capital or human resources to make big leaps"; many were weakened by the 737 MAX and Covid crises after a decade of price pressure.
- Unnamed supply chain executive: "They not only tell us what hardware we can buy, they are now going to specify all this fancy digital junk that goes on top of it?"
- Critics say Boeing has made similar bold pledges on a digital revolution "with mixed results."
- Richard Aboulafia (Teal Group): "Is it worth pursuing? By all means… Will it solve all their problems? No."
- Skeptics point to problems on the 777X and T-7A, both developed with digital tools. Industry sources said the technology must come with organizational and cultural change.
- Not found in sources: a count of critical suppliers (e.g., "~300"), supplier cost figures, or measured results after Dec. 2021.

Sources: Reuters, "Boeing wants to build its next airplane in the 'metaverse'" (Eric M. Johnson and Tim Hepher, Dec. 17, 2021), as syndicated by Investing.com, Euronews and TBS News.

Appendix A4: 777 FAUB Robots – What Was Built, What Failed, What Replaced It

What FAUB was and what went wrong

- Since 2015 Boeing built 777 fuselages upright, with robots drilling holes and installing fasteners, an initiative called fuselage automated upright build (FAUB) (Reuters).
- Robots worked in tandem to drill precisely and fasten upright metal panels; it was showcased as advanced manufacturing for the 777X and planned for future programs (Bloomberg).
- After “four years of trial and error,” Boeing dropped the robots for two main fuselage sections of the 777 and 777X (Bloomberg, Nov. 14, 2019).
- Boeing struggled to keep robots in sync inside and outside the panels, creating production snarls. A 2016 Seattle Times report described overtime and unfinished jobs completed after jets left the factory (cited by Bloomberg; not verified).
- Industry sources told Reuters the system caused reliability and rework problems; Boeing faced delays in 2018 when FAUB was tried on existing 777s.
- Jason Clark, VP of 777X production: “It was hard. It took years off my life.” He added: “It’s taught us how to design for automation.”

What replaced it

- Flex tracks: an automated system that drills holes around the fuselage circumference, after which mechanics fasten manually. Developed and honed on the 787.
- Rollout on the 777 fuselage began in Q2 2019; transition expected to finish by year-end 2019.
- Boeing spokesman Paul Bergman: “The flex track solution has proven more reliable, requiring less work by hand and less rework, than what the robots were capable of.”
- Parts of the build were redesigned to replace rivets with less difficult fasteners, improving ergonomics.
- No change planned in total 777 staffing at Everett, Washington.
- Boeing continues robotic systems elsewhere, such as 777X wing manufacturing.
- Not found in sources: FAUB investment, cost, or production-rate impact.

Appendix B5: 737 MAX Rate 47 and the Vaero Wing-Part Dispute

737 MAX rate increase to 47 per month

- May 27, 2026 (Bernstein conference): Ortberg said Boeing completed an FAA capstone review to go from 42 to 47 MAX per month “this summer.”
- History: output was capped at 38 per month after the Jan. 2024 door plug accident; the FAA lifted the cap in Oct. 2025, allowing 42.
- An FAA spokesperson said FAA inspectors “conducted extensive reviews of Boeing’s production lines to ensure this production rate increase will be done safely.” The article body does not describe a formal approval document; “approves” is the headline’s wording.
- Ortberg: “It’ll probably take us a few months of stabilisation there.” On 52 per month: “that’s going to be more of a strain than going to 47 because of the inventory.” Boeing raises rates in steps of five, about six months apart.

Vaero dispute (Reuters, Oct. 2026)

- Vaero, a small Florida supplier, is Boeing’s sole-source supplier of the trailing-edge rib kit, a wing component for the 767 and the current 777 Freighter. The composite-wing 777X is not affected.
- Boeing sued in Aug. 2026, saying Vaero stopped deliveries and stalled wing production in Everett; it called the part “indispensable” and cited costs of up to $6.5 million a day. The companies settled in late Sept. 2026.
- Reuters: Boeing “has not built any 777 freighters for nearly two months”; workers said production had not restarted after the settlement. The 767 line, including freighters and KC-46 tankers, was also affected.
- Deliveries: 21 777 Freighters so far in 2026 versus 35 in 2025; backlog 38 at end of September. Boeing is trying to make the kits in-house.
- Not confirmed: a search of Boeing’s Q3 2026 10-Q found no mention; the Leeham News article could not be read.

Sources: Seattle Times via AviationPros (May 29, 2026); CAPA (May 28, 2026); Reuters via STAT Times (Oct. 25, 2026).

Appendix B1: SPS Technologies Fire (Feb. 2025) – Sole-Source Fastener Exposure

What happened

- Fire on Feb. 17, 2025 at an SPS Technologies plant in Jenkintown, Pennsylvania, about 10 miles north of Philadelphia. Police reported an explosion in the building; a shelter-in-place order was issued; the fire took several days to extinguish; no workers were injured.
- WSJ (Mar. 6, 2025): the plant, about the size of 10 football fields, was “the sole supplier of some critical fasteners used in Boeing planes.”
- SPS is a unit of Berkshire Hathaway’s Precision Castparts. Analyst Kevin Michaels: “one of the country’s largest aerospace fastener plants, providing about 15% of the industry’s supply.”
- Other customers include GE, Safran and Airbus; GE and Safran supply engines for Boeing jets.
- Not confirmed in sources: the “40 parts” and “9+ months to re-source” figures, whether Boeing had any early warning, and any dollar loss.

Boeing impact and response

- CEO Kelly Ortberg: Boeing relies on SPS for many parts on both the 737 and 787. A parts shipment due out before the fire was inaccessible afterward.
- Boeing declined to say how many parts SPS alone provides or whether it could reach the blocked inventory.
- WSJ: finding alternative suppliers “isn’t an easy task”; the fasteners must meet the demands of air travel, some designs are complex, and they are used in jet engines, landing gear and other parts.
- Boeing said the loss would not have an immediate effect on production; suppliers and analysts expected fallout. Context: Boeing was targeting 38 MAX per month later in 2025 and rolled out about 35 in February (analyst estimate).
- Ortberg: “This is an example of a hiccup in the supply chain.”

Appendix B2: Alaska Airlines Flight 1282 – Missing Bolts and Records

The event and how it happened

- Jan. 5, 2024, about 1713 PST: Boeing 737-9 N704AL climbing from Portland (PDX) to Ontario, California, about six minutes after takeoff. The left mid exit door (MED) plug separated near 14,830 ft; the crew returned and landed. One flight attendant and 7 passengers had minor injuries; 177 people were aboard.
- NTSB: “the four bolts that secured the left MED plug to prevent it from moving upward vertically were missing before the newly manufactured airplane was delivered to Alaska Airlines.”
- The fuselage was built at Spirit AeroSystems in Wichita, which closed the plug July 28, 2023. At Boeing’s Renton plant, arrival inspection found improperly installed rivets; Boeing personnel opened the plug to rework them (Sept. 2023).
- NTSB: no personnel experienced with opening or closing an MED plug were on duty at the times, and none said they knew who opened it.

Records, probable cause and recommendations

- Boeing’s parts-removal procedure required a removal record to open the plug. NTSB: “no removal record was generated”; the plug was closed without its bolts and “no quality assurance inspection of the plug closure was performed.”
- The tracking system was Boeing’s Common Manufacturing Execution System (CMES), a web-based system for instructions, nonconformances and signoffs; no removal record was found in it.
- Probable cause: Boeing’s failure to provide adequate training, guidance and oversight for its parts removal process. Contributing: the FAA’s ineffective compliance enforcement surveillance and audit planning.
- The procedure had a documented history of compliance issues for at least 10 years. Board meeting June 24, 2025 (report AIR-25-04); 19 new recommendations, 11 to the FAA and 8 to Boeing.
- Not confirmed: who opened the plug or the exact date.

Sources: NTSB investigation DCA24MA063 and report AIR-25-04; Airline Geeks and AVweb (July 10, 2025); Jalopnik (June 26, 2025).

Appendix B3: Supplier Readiness and Visibility Survey Evidence

Ramp-up readiness (McKinsey and Aviation Week, July 2024)

- Survey of 175 leaders across more than 80 firms (airframe and engine OEMs, suppliers, MROs, operators), run by McKinsey with Aviation Week for Farnborough 2024.
- Finding: “65% of suppliers indicated that their current production systems are inadequate to reach airframer ramp-up targets.” The denominator is not stated.
- “The shortfall appears to be most pronounced at engine and Tier 2 suppliers” (hedged wording in the article).
- McKinsey partner Henry Marcil: “Four-fifths of suppliers say they are not ready to deal with it.”
- Gaps cited: production plant and equipment capacity and shortage of skilled labor, with no percentages given.

Roland Berger European surveys and visibility

- 2025 report (with BDLI, GIFAS, ADS; 130 small and medium suppliers): nearly 70% feel well prepared for ramp-up; half need more financial resources; two-thirds still face disruptions; “Tier-2 and Tier-3+ suppliers are particularly affected.”
- The same report: 65% cited personnel shortages as the most common challenge. This is a different metric from the McKinsey 65%.
- 2026 survey (95 European suppliers and OEMs): 56% struggle to get the right people in the right jobs; more than two-thirds say they are well placed for ramp-up.
- 2023 survey (105 German aerospace and defense participants): supplier data integration was a mid-term priority for 12%; transparency was discussed qualitatively only.
- Not found in any source read: “12% of procurement teams can see more than half of their Tier 2 suppliers.” The nearby 12% is the data integration figure above.

Sources: Aviation Week, Christine Boynton and Alan Dron (July 24, 2024); Roland Berger press release (July 4, 2025), articles of July 3, 2025, May 31, 2026 and Sept. 8, 2023.

Our Model: Annual Cost by Component ($M)

Component	2027	2028	2029	2030	2031	5-yr total
Workforce nonprofit	7.2	6.8	6.1	6.1	6.1	32.3
Partnerships & supply access	0.0	14.2	8.0	5.5	2.0	29.7
3D printing + FAA	0.0	1.0	12.0	9.5	6.5	29.0
Digital twin	0.0	1.0	3.5	3.0	1.5	9.0
Total	7.2	23.0	29.6	24.1	16.1	100.0

Source: team cost model (Cost of the Strategy slide). Component totals are our arithmetic from the chart data.

Our Model: Key Assumptions

Item	Assumption	Where it shows up
Training	$12K per trainee	Workforce nonprofit line; Phase 1 tuition budget is $500K for 50 trainees ($10K each)
Partnerships	$18M co-investment	Partnerships & supply access line ($29.7M total over five years)
3D printing	~$1.5M per metal printer	3D printing + FAA line ($29.0M total)
Digital twin	$6M to build	Digital twin line ($9.0M total, so ~$3.0M of operating cost)
Benefits	40% of spend re-sourced	Drives net 5-year value of +$9.8M and 1.1x return

Our Model: Headline Results

Metric	Value	How to explain it
Total investment	≈$100M	Sum of annual costs 2027–2031 ($7.2M + $23.0M + $29.6M + $24.1M + $16.1M)
Net 5-year value	+$9.8M	Implies roughly $109.8M of benefits against $100M of cost
Return on investment	1.1x	$109.8M / $100M = 1.098x
Peak cumulative outlay	≈$41M	Less than the $59.8M gross cost through 2029, so we describe it as net of benefits. Confirm definition with the model owner.
Break-even	2031	Cumulative benefits catch up with cumulative cost in the final year

Our Model: Scale of the Plan vs Boeing Cash

Share of Boeing's 2026 free cash flow outlook ($1–3B)

$20M	0.7–2.0%	$29.6M	1.0–3.0%	3.3–10%
Average annual cost ($100M / 5 years)	Average year as a share of the $3B to $1B outlook	Peak year (2029)	Peak year as a share of the $3B to $1B outlook	Whole five-year plan as a share of one year's outlook

Source: our arithmetic from the team cost model and Boeing Case Booklet, Exhibit 1.

Our Model: Cost vs Acquiring Capacity

Alternative	Price	Multiple of our $100M plan
Arcline acquires Kaman (2024)	~$1.8B	~18x
Arcline acquires Novaria (2025)	$2.2B	~22x
GE Aerospace agrees to buy CPP (Sept. 2026)	$11.75B	~118x
Boeing completes Spirit AeroSystems (Dec. 2025)	~$8.3B incl. ~$4B debt	~83x

Sources: Case Booklet Exhibits 1 and 7; Leeham News (Sept. 9, 2026) for GE/CPP; Manufacturing Dive (Dec. 9, 2025) for Boeing/Spirit. Multiples are our arithmetic.

Our Model: Cost of a Stoppage vs Our Plan

Reference point	Figure	Comparison
Vaero dispute: cost cited by Boeing	Up to $6.5M per day	About 15 days of a Vaero-type stoppage ≈ the $100M plan
Same cost over 30 days	$195M	About 2x the five-year plan
Deliveries hit	21 777 Freighters in 2026 vs 35 in 2025	Shows how one sole-source part idles a line

Sources: Reuters via STAT Times (Oct. 25, 2026), as summarized in Appendix B5 of the main deck; our arithmetic. Boeing's own cost claim, not an audited figure.

Our Model: Phases 2 and 3 Recap

BOEING

Phase	Initiative	What it does
Phase 2 (Year 2)	Workforce partnerships	Connect Phase 1 trainees to supplier-confirmed entry-level machine operator openings
Phase 2 (Year 2)	Production agreements	Multiyear purchase commitments for reserved capacity, delivery milestones and supplier reporting
Phase 3 (Years 3–5)	Supply chain digital twin	Map critical Tier 2/3 suppliers to predict disruptions and guide investments
Phase 3 (Years 3–5)	Qualify 3D-printed alternatives	Develop certified backup components to reduce sole-source dependency

Our Model: Timeline Recap

Workstream	Start	End
501(c)(3) setup & governance	2027 Q1	2027 Q2
Curriculum, employer board & training lab	2027 Q1	2027 Q3
Training cohorts & Tier 2/3 placement	2027 Q3	2030+
Supplier partnership network	2027 Q4	2028 Q2
Foundry & supply access agreements	2028 Q1	2028 Q4
3D printing cell build & operation	2028 Q4	2029 Q3
FAA qualification & part approvals	2029 Q1	2030+
Digital twin analytics system	2028 Q3	2029 Q2

Source: read from the Timeline slide bars; confirm quarter boundaries against the team plan.

Our Model: KPI Recap

KPI	Definition	Baseline	Target	Timing
Workforce Development	Students enrolled in Boeing Ed	750 workers	1,000 trained	Year 1
Supplier Partnerships	New capacity agreements secured	2 agreements	3 agreements	Year 2
Alternative Sourcing	Backup part numbers qualified	1 part	2 parts	Year 5
Supply Chain Visibility	Priority Tier 2/3 suppliers mapped	70% mapped	90% mapped	Year 5

Our Model: Governance Recap

Workstream	Owner	Forum
Nonprofit	Boeing program director/board	Monthly nonprofit supplier review; approve cohorts against confirmed hiring needs
Supplier production partnerships	Boeing procurement lead	Quarterly supplier steering committee; approve commitments, resolve delivery issues
3D-printing partnerships	Boeing additive manufacturing lead	Engineering and quality stage reviews; approve part selection, testing, production release
Digital twin implementation	Boeing manufacturing engineering lead	Monthly engineering supplier review; validate the model before operational changes

Our Model: Risk Recap

Risk	Likelihood	Impact	Mitigation
Workforce placement	High	Medium	Target verified vacancies; offer voluntary supplier recruiting pathways
Supplier commitment	Medium	High	Multiyear orders; reserved capacity with ownership-change protections
3D-printing qualification	High	High	Prioritize feasible parts; phased testing and qualification milestones
Digital twin data quality	High	High	Agreed data sharing; validate supplier inputs; pilot before scaling

Reconcile Before Q&A

Item	What a judge may notice	Suggested handling
KPI timing vs trainee count	Workforce KPI target reads "1,000 trained, Year 1", but Financial Impact plans 1,000 trainees over five years and Phase 1 trains 50 pilots.	Restate as cumulative 1,000 by Year 5, or 50 in Year 1.
KPI baseline of 750 workers	Boeing Ed does not exist yet, so what the 750 baseline counts is unclear.	Define it (for example, current supplier trainees in the target trades) or remove it.
$12K vs $10K per trainee	Cost assumptions use $12K per trainee; Phase 1 and Financial Impact use $10K tuition.	Say $10K is tuition and $12K includes placement support, or align the numbers.
Workforce line is $32.3M, tuition is $10M	About $22M of the workforce line is not tuition.	Be ready to name it: instructors, equipment, lab, administration, supplier mentors.

Reconcile Before Q&A

BOEING

Item	What a judge may notice	Suggested handling
Peak cumulative outlay ≈$41M	Gross cumulative cost through 2029 is $59.8M, so $41M must be net of benefits.	Define it on the slide or in the answer.
Digital twin: $6M to build vs $9.0M total	The extra ~$3.0M must be operating cost.	Say so explicitly.
Strategy Overview has no name	The main deck still shows "[Strategy Name]" in its source; the standardized deck just says Strategy Overview.	Pick a name for the strategy before presenting.
Supplier agreements KPI looks small	Target moves from 2 to 3 agreements in Year 2.	Justify: agreements are with critical suppliers only, each with reserved capacity and delivery milestones.

Evidence: FAA Approach to 3D-Printed Parts

BOEING

How the FAA treats additive manufacturing

- AM certification fits within the existing FAA regulatory system and is treated like other process-intensive material systems (FAA, 2020).
- The approach is risk-based; higher-risk uses go through Aircraft Certification standards branches, and maintenance spare-part requests are handled case by case.
- The FAA intends performance-based policy that lets applicants show compliance through consensus standards (SAE, ASTM/ISO F42, AWS D20).
- The 2017 roadmap described three pathways: point-design certification, part-family qualification and model-enabled qualification.
- No duration for FAA certification is stated in the sources we read.

Sources: FAA Report to Congress on Additive Manufacturing (Mar. 31, 2020); FAA AM Roadmap briefing to REDAC (Mar. 2017); FAA AM R&D status (Aug. 2022).

Evidence: Cost, Time and Barriers of AM Qualification

BOEING

Why qualification is the long pole

- NIST/NASA/FAA strategy document (Mar. 2026): comprehensive allowables for one material and process cost more than $1M and take more than 18 months; each new machine, powder, build practice or part still needs significant qualification effort.
- Barriers: qualification is largely empirical, regulators and OEMs do not yet trust computational models, microstructure varies by location, and validation data is mostly proprietary.
- FAA: inspection struggles with complex geometry and rough as-built surfaces; AM-specific defects may go undetected.
- Recommended starting point: Ti-6Al-4V and Alloy 718 (laser powder bed), staged off-ramps, and delta qualification of machines.

Sources: NIST CM4QC Strategy Document, NASA/TM-20260001729, DOT/FAA/TC-26/10 (Mar. 4, 2026); FAA Report to Congress (2020); FAA AM R&D status (2022).

Evidence: AM and Mechanical-Standards Parts

BOEING

What the sources do and do not show

- Confirmed in sources: a printed titanium bracket on a series Airbus A350 (2017) and structural titanium parts on the 787.
- Not confirmed: any commercial-aircraft use of printed bushings, fasteners or bearings. Treat these as unproven.
- FAA expects safety-critical metal AM airframe and engine parts "in the near future" (2020), but fatigue-critical, long-life parts face demanding testing.
- Implication for our pitch: lead with brackets and fittings-type parts and say plainly that bearings depend on partnerships and training.
- No credible machine price range was found; the $1.5M per metal printer is a team assumption.

Sources: Airframer (Sept. 19, 2017); FAA Report to Congress (2020); Materialise (undated, 2026). Machine price: not confirmed.

Evidence: Qualification Time for New Sources

What we could and could not confirm

- The booklet says qualifying a replacement source takes years and a brand-new source takes longer than a typical lead time.
- Norsk Titanium (M4S/TCT, June 1, 2026): customer-side validation of a new additive supplier typically takes "weeks or months"; Nadcap accreditation is expected to remove weeks or months before first delivery.
- No credible duration or cost figures were found for AS9100, first article inspection, or Boeing specification approvals such as D1-4426.
- Use the booklet's "years" language rather than inventing a number.

Sources: Boeing Case Booklet (p. 7); M4S/TCT Magazine (June 1, 2026).

Evidence: GE Aerospace Buying CPP

The castings example from the booklet

- GE announced the deal on Sept. 8, 2026 at $11.75B ($7B cash, rest debt); sellers are Warburg Pincus and Berkshire Partners.
- CPP is an Ohio castings maker supporting LEAP, GEnx and defense programs; it would run as a separate unit and keep serving existing customers, including GE competitors.
- Expected close is the second half of 2027, pending regulatory approval. Current status as of Nov. 2026 is not confirmed.
- GE expects airfoil demand to rise about 30% by 2030.

Source: Leeham News (Sept. 9, 2026). Another outlet's headline cites 26x 2027 EBITDA; we did not read it.

Evidence: Cost, Time and Barriers of AM Qualification

BOEING

Why qualification is the long pole

- NIST/NASA/FAA strategy document (Mar. 2026): comprehensive allowables for one material and process cost more than $1M and take more than 18 months; each new machine, powder, build practice or part still needs significant qualification effort.
- Barriers: qualification is largely empirical, regulators and OEMs do not yet trust computational models, microstructure varies by location, and validation data is mostly proprietary.
- FAA: inspection struggles with complex geometry and rough as-built surfaces; AM-specific defects may go undetected.
- Recommended starting point: Ti-6Al-4V and Alloy 718 (laser powder bed), staged off-ramps, and delta qualification of machines.

Sources: NIST CM4QC Strategy Document, NASA/TM-20260001729, DOT/FAA/TC-26/10 (Mar. 4, 2026); FAA Report to Congress (2020); FAA AM R&D status (2022).

Evidence: GE Aerospace Buying CPP

The castings example from the booklet

- GE announced the deal on Sept. 8, 2026 at $11.75B ($7B cash, rest debt); sellers are Warburg Pincus and Berkshire Partners.
- CPP is an Ohio castings maker supporting LEAP, GEnx and defense programs; it would run as a separate unit and keep serving existing customers, including GE competitors.
- Expected close is the second half of 2027, pending regulatory approval. Current status as of Nov. 2026 is not confirmed.
- GE expects airfoil demand to rise about 30% by 2030.

Source: Leeham News (Sept. 9, 2026). Another outlet's headline cites 26x 2027 EBITDA; we did not read it.

Evidence: Sole-Source Pricing Power

TransDigm and the DoD Inspector General

- 2019 report: excess profit of about $16.1M on 46 parts (about $26.2M of sales, 2015–17); 46 of 47 sampled parts showed excess profit; 39 of 47 were made only by TransDigm.
- 2021 report: $20.8M of excess profit on 105 spare parts across 150 contracts; the IG noted audits over 23 years highlighting excess profit on sole-source contracts.
- TransDigm disputed the method, arguing profit should be measured against revenue rather than cost, and refunded $16M in 2019 without admitting wrongdoing.
- Use carefully: this is defense spares buying, not Boeing commercial supply, but it shows what sole-source status can do to pricing.

Sources: Inside Defense (Feb. 27, 2019); Defense News (Dec. 14, 2021).

Evidence: Antitrust Context

What regulators have said and done

- No FTC or DOJ action on aerospace PE roll-ups was found.
- Closest precedent read: the FTC's Jan. 2025 consent order with Welsh Carson over the U.S. Anesthesia Partners roll-up (healthcare), alleging serial acquisitions let practices raise prices.
- Remedies there included prior-approval requirements for future deals; commissioners called it an example of enforcement against serial acquisitions.
- Our plan does not acquire suppliers, so it avoids this exposure; a selective acquisition would need antitrust review, as Boeing/Spirit showed.

Sources: Duane Morris Antitrust Law blog (Jan. 17, 2025); FTC press release (Feb. 2026).

Evidence: Customer-of-Choice Concept

What the research says

- Preferred customer status: a buyer is "awarded preferential treatment" by its important suppliers compared with their other customers (Schiele, Veldman and Hüttinger, 2011).
- Proposed drivers: customer attractiveness, supplier satisfaction, and the supplier’s knowledge of alternative customers.
- An undated abstract reports that preferred customers get early access to innovation and talent, which matters most when supply is scarce.
- Not confirmed: a source ranking payment reliability, volume visibility and technical support. Present those as our hypotheses.

Sources: IMP Group page (https://impgroup.org/?p=5288); CORE abstract (core.ac.uk/works/73077427), authors not shown.

Evidence: Payment Terms and Supplier Finance

How terms and early payment work

- Boeing extended supplier payment terms from 30 days to up to 120 days; most aerospace makers pay on net 60, 90 or 120 (Global Trade Review, circa 2016).
- Early payment through supply-chain finance depends on a discount fee and lender program eligibility.
- A Boeing / U.S. Exim / Citibank program had $740M initial capacity, with Exim guaranteeing up to 90% of risk; eligibility rules limited it to export-related receivables and U.S. content (undated article).
- Exhibit 5 in the booklet shows why prepaid buyers beat 120-day terms for a bearing supplier.

Sources: Global Trade Review articles (undated, circa 2012–2016); Case Booklet Exhibit 5. Current programs not confirmed.

Evidence: Long-Term Supplier Agreements

BOEING

Precedent for multiyear commitments

- Pratt & Whitney signed more than 135 long-term agreements with projected spend of $16B (Oct. 2014), so suppliers would invest in equipment, people and training for a planned doubling of engine production.
- American Machinist reported these as intended benefits only; no measured results, contract terms or delivery performance were found.
- Treat as a precedent that the structure exists, not as proof it works.

Sources: PR Newswire (Oct. 2, 2014); American Machinist (Oct. 3, 2014).

Evidence: Machinist Wages and Demand

BLS data (Occupational Outlook Handbook, Oct. 29, 2026)

- May 2025 median pay: machinists $58,750; tool and die makers $64,050; combined $59,620 ($28.66 per hour).
- Employment outlook 2025–35: machinists +1%; machinists and tool and die makers combined -1%.
- About 30,400 openings per year, mostly from workers leaving the occupation.
- Typical entry: high school diploma plus on-the-job training; BLS has no separate CNC operator profile.
- Implication: demand is steady replacement hiring, not growth, which supports targeting verified vacancies.

Source: U.S. Bureau of Labor Statistics, Machinists and Tool and Die Makers.